BPK 207 Study Guide: Sensorimotor Contribution Part 2 — The Vestibular System (Chapter 3b, Slides 1)

Course: BPK 207 • **Session** / **Chapter:** 3b

> **Big-Picture Synthesis**
>
> Chapter 3b explores the sensory contributions of the vestibular and visual systems to sensorimotor control, balance, and navigation. This section focuses specifically on the vestibular system, detailing its dual role in detecting linear and angular head acceleration through specialized end organs (otoliths and semicircular canals). By utilizing bilateral paired apparatuses, the vestibular system achieves a high signal-to-noise ratio and precise spatial orientation across multiple planes of motion (roll, pitch, and yaw).

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1. Vestibular System Anatomy & Functional Overview

The **vestibular system** is essential for detecting head acceleration, determining

head position and motion, and maintaining body orientation and balance. It relies on bilateral end organs housed within the inner ear on both sides of the head.

- **Bilateral Function**: Vestibular apparatuses on both sides of the head work together in a push-pull arrangement. This bilateral integration increases the **signal-to-noise ratio**, greatly enhancing sensitivity to subtle motion.
- **End Organ Components**:
- **Otoliths**: Comprising the **saccule** and **utricle**, these structures detect **linear head acceleration** (translational movement through the environment) and static changes in head orientation relative to gravity.
- **Semicircular Canals**: Comprising the **superior**, **posterior**, and **horizontal canals**, these structures detect **angular head acceleration** (turning, tilting, and rotatory body movements).

Head movement in the natural environment is rarely purely linear or rotational; rather, it is a complex combination across various planes. The vestibular system resolves this using three principal rotational dimensions: **roll** (rotation around the x-axis), **pitch** (rotation around the y-axis), and **yaw** (rotation around the z-axis).

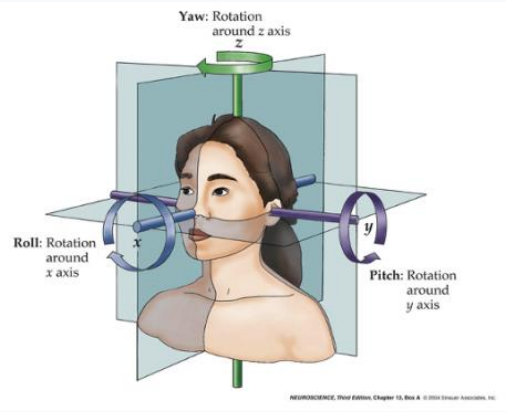


Diagram Walkthrough (slide_01_fig_1.jpeg): Overview of the vestibular apparatus (top figure) showing the otolith organs (utricle and saccule) and the three semicircular canals (superior, posterior, and horizontal). The bottom-right figure

illustrates rotational planes of the head: roll (x-axis), pitch (y-axis), and yaw (z-axis).

> **High-Yield Exam Takeaways & Common Traps**
>
> - Be certain to distinguish linear acceleration (otoliths: utricle and saccule) from angular acceleration (semicircular canals).
> - Bilateral vestibular organs work synergistically to maximize the signal-to-noise ratio for motion detection.
> - Memorize the three rotational axes: roll (x), pitch (y), and yaw (z).

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2. Otolith Organs & Semicircular Canal Specializations

To accurately encode spatial kinematics, the vestibular end organs divide labor based on the physical nature of the acceleration forces acting upon the head.

Otolith Mechanics & Planes

- **Saccule**: Oriented primarily in the **vertical plane**, making it sensitive to vertical linear acceleration (e.g., riding an elevator up or down).
- **Utricle**: Oriented primarily in the **horizontal plane**, making it sensitive to horizontal linear acceleration (e.g., accelerating in a car) as well as static **head tilt** relative to gravity.

Semicircular Canal Mechanics

- **Angular Specialization**: The three semicircular canals are oriented

orthogonally (at roughly 90-degree angles to one another) to capture rotational velocity across three-dimensional space.

- **Inertial Coupling**: Movement of the head in any rotational direction causes fluid within the corresponding canal to flow relative to the skull, deflecting sensory hair cells and initiating downstream neural coding.

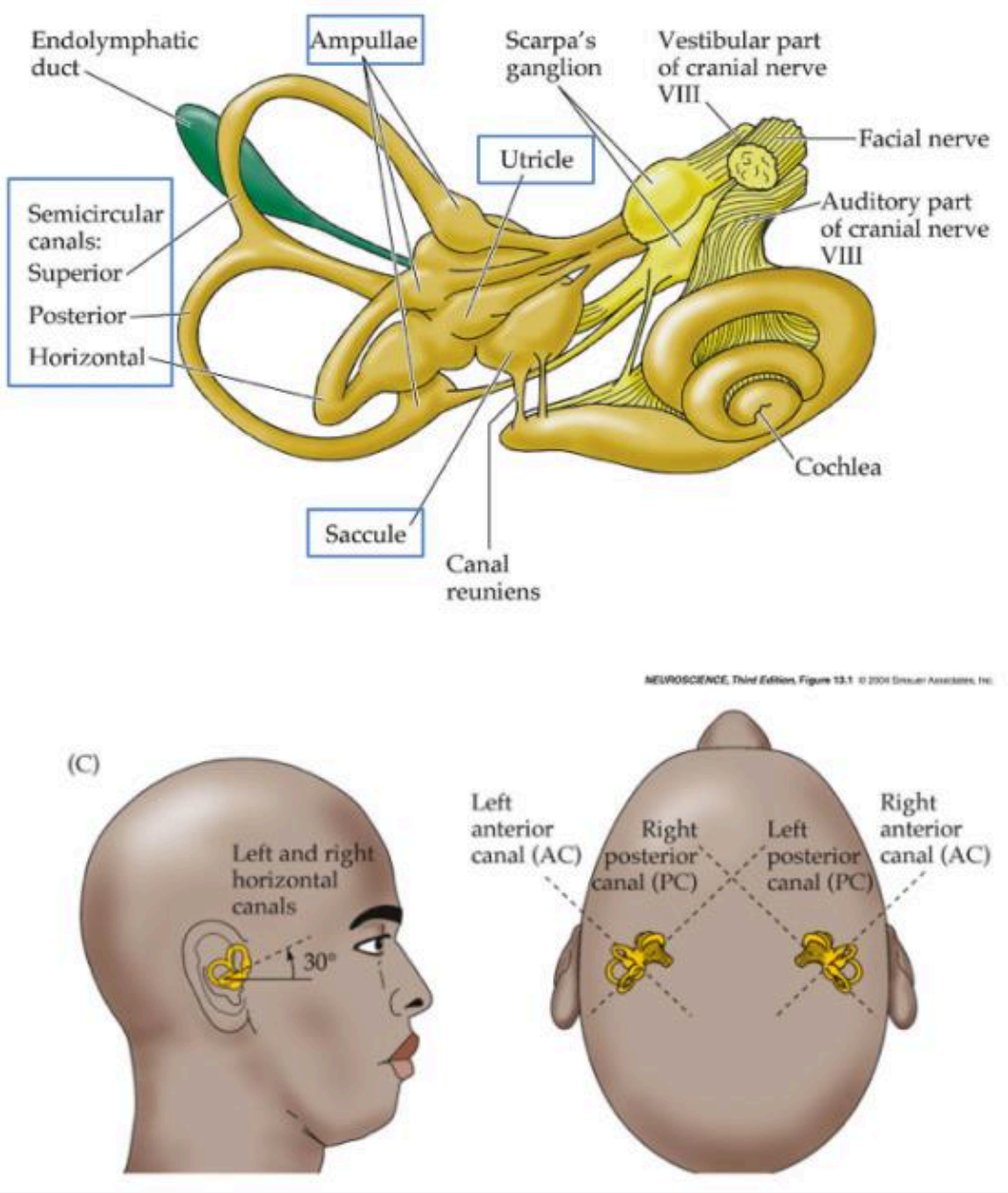


Diagram Walkthrough (slide_01_fig_2.jpeg): Detailed anatomical alignment of the semicircular canals and otolith organs, highlighting their orthogonal spatial orientation and innervation pathways via the vestibular branch of cranial

nerve VIII.

High-Yield Exam Takeaways & Common Traps

- Trap alert: Do not confuse the planes of the otoliths. The utricle senses horizontal acceleration and head tilt, whereas the saccule senses vertical acceleration.
- Semicircular canals are exquisitely tuned to detect angular velocity and rotational acceleration through orthogonal spatial arrangement.

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3. Vestibular Hair Cell Transduction: From Mechanical Deflection to Electrical Signals

Vestibular transduction converts mechanical head motion into electrical signals through specialized **hair cells** embedded in gelatinous matrices. Each hair bundle features a tallest projection called the **kinocilium**, alongside shorter stereocilia connected by tiny extracellular filaments known as **tip links**.

Step-by-Step Transduction Cascade:

1. **Resting State (Upright Head):** When the head is upright and stationary, tip links maintain a slight baseline tension. This keeps spring-gated ion channels partially open, allowing a small influx of **K^+** from the surrounding K^+-rich endolymph. This creates a tonic **resting discharge** in primary vestibular afferents.

2. **Depression** / **Excitation (Toward Kinocilium):** When stereocilia are mechanically bent **toward the kinocilium**, tip links are stretched further,

opening more ion channels. **K^+ influx** depolarizes the hair cell, triggering voltage-gated Ca^{2+} channels, neurotransmitter release, and an **increased afferent firing frequency**.

3. **Hyperpolarization / Inhibition (Away from Kinocilium):** When stereocilia are bent **away from the kinocilium**, tip link tension relaxes, closing the ion channels. The cell hyperpolarizes, and afferent discharge decreases or stops entirely.

Bending Direction	Tip Link Tension	Ion Channel State	Membrane Potential	Afferen Firing Rate
Toward Kinocilium	Increased	More Open	Depolarization	Increase (Excitati
Away from Kinocilium	Decreased	Closed	Hyperpolarization	Decreas (Inhibitic

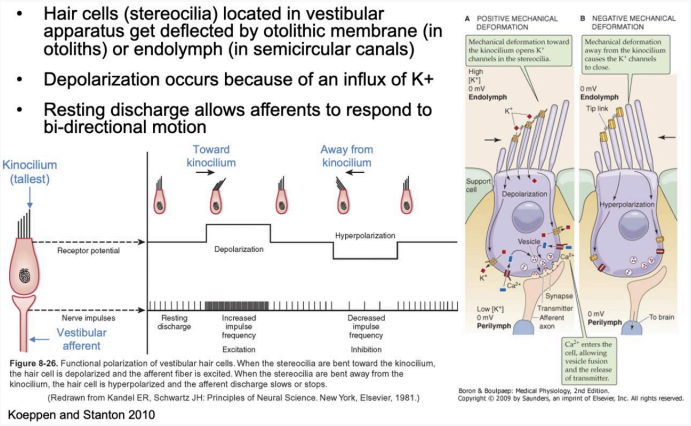


Figure 8-26. Functional polarization of vestibular hair cells. When the stereocilia are bent toward the kinocilium, the hair cell is depolarized and the afferent fiber is excited. When the stereocilia are bent away from the kinocilium, the hair cell is hyperpolarized and the afferent discharge slows or stops.
(Redrawn from Kandel ER, Schwartz JH: Principles of Neural Science. New York, Elsevier, 1981.)
Koeppen and Stanton 2010

Diagram Walkthrough (slide_02_fig_1.jpeg): Observe the structural polarization of vestibular hair cells. Panel A shows mechanical deformation toward the kinocilium stretching tip links, opening K+ channels, and causing depolarization.

Panel B shows deformation away from the kinocilium closing channels and causing hyperpolarization.

High-Yield Exam Takeaways & Common Traps

- Resting discharge is essential because it allows vestibular afferents to signal bi-directional motion (both increases and decreases in firing rate).
- Unlike auditory hair cells bathed in perilymph, vestibular stereocilia project into K^+-rich endolymph, driving K^+ influx upon channel opening.
- Tip links act as mechanical spring gates linked directly to ion channels.

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4. Semicircular Canals: Sensing Angular Acceleration via Endolymph Inertia

The three semicircular canals (superior, posterior, and horizontal) detect rotational head movements across the three spatial dimensions (**roll around x-axis, pitch around y-axis, and yaw around z-axis**).

The Mechanism of Angular Acceleration Sensing:

1. **Endolymph Inertia:** The canals are filled with a viscous, K^+-rich fluid called **endolymph**. When the head rotates, inertia causes the endolymph to lag behind the physical movement of the skull (analogous to coffee lagging behind a rotated mug).

2. **Cupula Displacement:** This relative fluid flow pushes against a gelatinous barrier called the **cupula**, which spans the ampulla of the canal.

3. **Hair Cell Deflection:** Embedded within the cupula are the hair cells. As the cupula distends under endolymph pressure, the hair bundles are bent, altering their firing rates to signal the velocity and direction of head rotation to the brainstem.

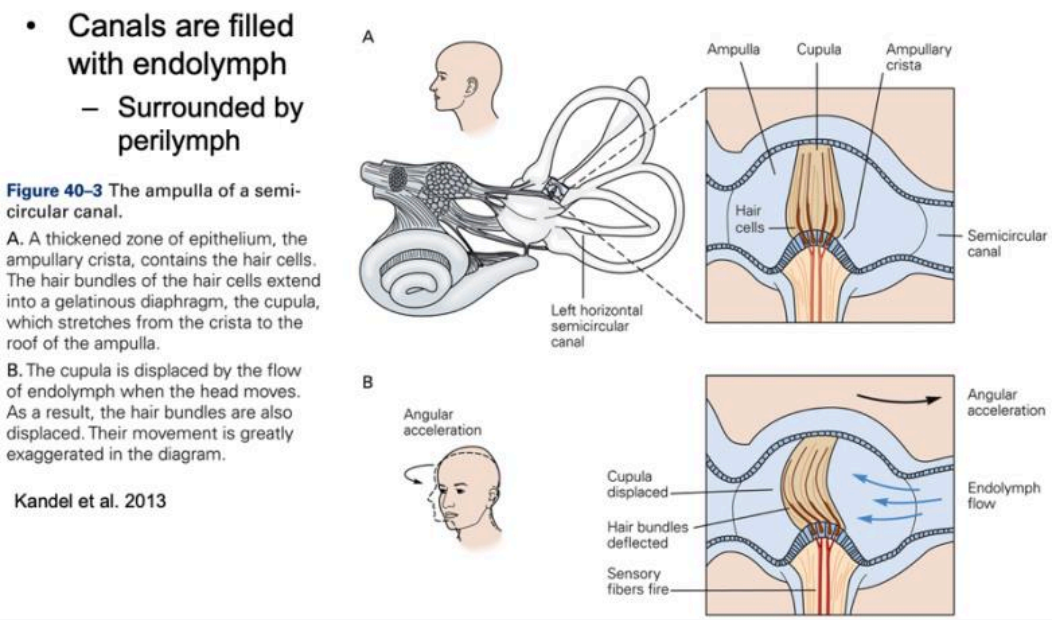


Diagram Walkthrough (slide_02_fig_2.jpeg): Examine the ampulla of the semicircular canal. Notice how endolymph flow displaces the gelatinous cupula, bending the embedded hair cell bundles and generating neural signals of angular acceleration.

High-Yield Exam Takeaways & Common Traps

- Endolymph moves relative to the head due to inertia; from an inertial frame of reference, the fluid stays still while the head and canals rotate around it.
- Pairs of canals on opposite sides of the head work push-pull style to maximize the signal-to-noise ratio during head rotation.

5. Otolith Organs: Sensing Linear Acceleration and Gravity

The otolith organs—consisting of the **saccule** and the **utricle**—detect linear (translational) head acceleration and static head orientation relative to gravity.

The Otolithic Membrane Mechanism:

1. **Otoconia Stones:** The otolithic membrane is a gelatinous layer weighted down by calcium carbonate crystals called **otoconia**.

2. **Inertial Lag:** When you accelerate linearly (or tilt your head), the heavy otoconia and otolithic membrane lag behind the underlying macula due to inertia (imagine sliding a cookie on a moving plate).

3. **Shearing Force:** This lag deflects the hair cells projecting upward into the membrane, signaling changes in translational velocity or gravitational vector.

Functional Division of Otolith Organs:

- **Utricle:** Senses horizontal linear acceleration (e.g., accelerating in a car) and head tilt relative to gravity.
- **Saccule:** Senses vertical linear acceleration (e.g., riding an elevator).

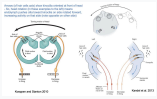


Diagram Walkthrough (slide_03_fig_1.jpeg): Review the otolith macula structure. Notice how the otolithic membrane containing otoconia stones lags behind head motion, shearing the underlying hair cells to signal linear acceleration

or gravitational tilt.

High-Yield Exam Takeaways & Common Traps

- The utricle operates in the horizontal plane, whereas the saccule operates in the vertical plane.
- Both otolith organs rely on an inertial mass (otolithic membrane with otoconia) that shifts relative to sensory hair cells during translational motion.

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6. Functional Roles of the Vestibular System

The vestibular system integrates inertial information from both the otoliths and semicircular canals to execute four primary functions critical for sensorimotor control: **postural stabilization**, **gaze stabilization**, **perception of self-motion**, and **spatial navigation**.

1. **Postural Stabilization**: Vestibular afferents project centrally to the **vestibular nuclei** in the brainstem, which subsequently give rise to descending **vestibulospinal tracts** that activate spinal motoneurons and postural muscles. This neural loop helps keep the head upright and maintains balance.

2. **Gaze Stabilization**: Operating in tight conjunction with the visual system, the **vestibulo-ocular reflex (VOR)** compensates for active head movements by rotating the eyes within their orbits in the opposite direction, preserving visual fixation on a target of interest.

3. **Perception of Self-Motion & Spatial Navigation**: Vestibular input informs the central nervous system (CNS) about body translation and rotation, updating internal models of spatial orientation and navigation.

> **High-Yield Exam Takeaways & Common Traps**
>
> - Vestibulospinal reflexes are essential for maintaining upright head posture and preventing falls during unexpected balance perturbations.
> - The VOR operates at extremely high speeds (~10 ms latency) to stabilize images on the retina during head turns.

7. Galvanic Vestibular Stimulation (GVS) as an Experimental Tool

Galvanic Vestibular Stimulation (GVS) is a non-invasive neurostimulation technique where weak direct electrical current is applied through electrodes placed over the mastoid bones directly behind the ears (comparable to tDCS).

- **Mechanism**: GVS depolarizes and hyperpolarizes vestibular afferents and hair cells across both the otoliths and semicircular canals.
- **Perceptual & Motor Output**: This artificial stimulation triggers an illusory perception of head and body tilt, inducing a stereotyped **compensatory sway response** in the opposite direction.
- **Directional Rule**: Anode-right GVS causes an illusory sensation of tilt toward the cathode (left), prompting the body to generate a compensatory sway toward the **anode** (right).
- **Postural Attenuation**: Sway responses are significantly attenuated when standing with a wider base of support (feet apart), demonstrating how the CNS dynamically reweights sensory inputs.

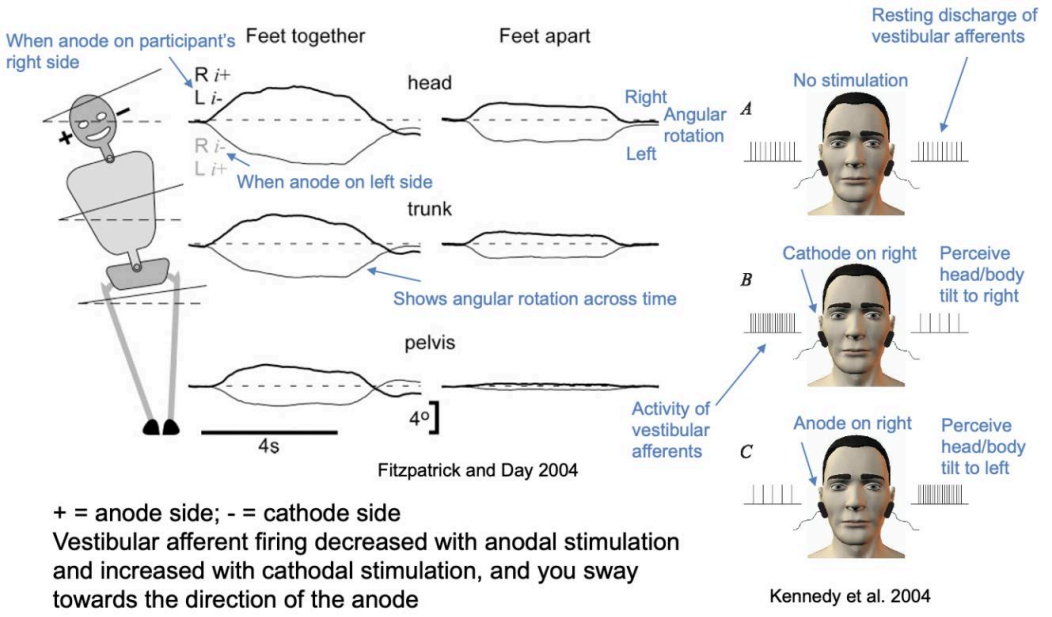


Diagram Walkthrough (slide_05_fig_1.jpeg): Slide 5 data showing how GVS delivered via mastoid electrodes induces directional body sway. Note that participants sway toward the anodal side to compensate for the illusory vestibular signal.

High-Yield Exam Takeaways & Common Traps

- Exam Trap: Students often confuse the direction of illusory sway versus compensatory sway. The participant feels sway toward the cathode and actively sways toward the anode (+).
- GVS provides direct experimental evidence that vestibular signals independently drive automatic postural adjustments without requiring natural head motion.

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8. Anatomy of the Eye and Optical Refraction

The primary function of the eye is to focus visual images onto the retina with minimal optical distortion. **Light rays** enter the eye and undergo **refraction** (bending) to converge at the retina.

Refraction occurs across two primary structures:

1. **Cornea**: Responsible for **2/3** of the eye's total optical refraction. Its curvature is fixed.

2. **Lens**: Responsible for the remaining **1/3** of total refraction. Unlike the cornea, the lens is flexible and can alter its shape and curvature to adjust the focal length for objects at varying distances.

This dynamic adjustment of the lens shape is known as **accommodation**, which is driven by the contraction and relaxation of the **ciliary muscles**. Note that the image projected onto the retina is inverted, requiring central nervous system interpretation to perceive orientation correctly.

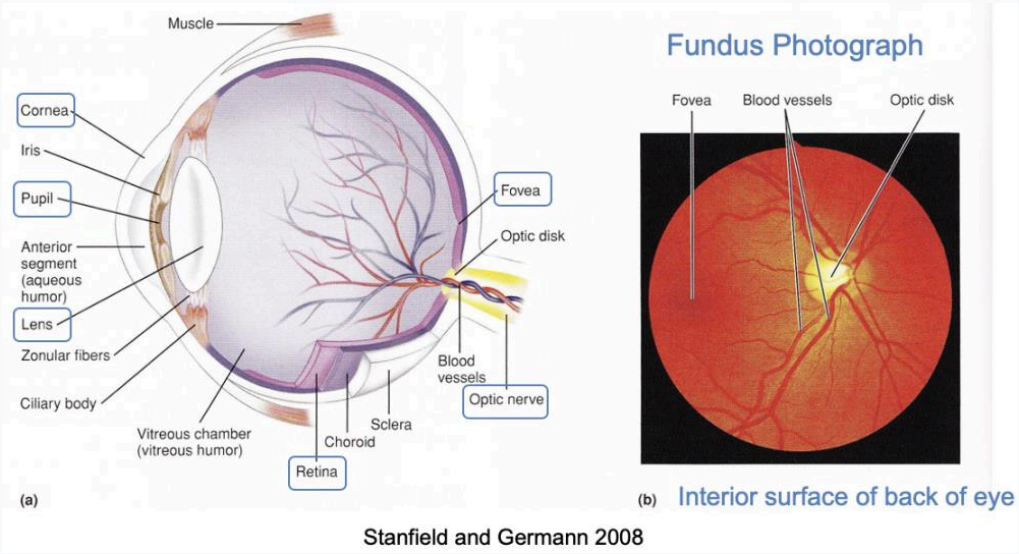


Diagram Walkthrough (slide_06_fig_1.jpeg): Anatomical cross-section of the eye showing key structures (cornea, pupil, iris, lens, zonular fibers, ciliary body, fovea, optic disk, and retina) alongside the fundus photograph and ray

diagrams illustrating corneal/lens refraction and image inversion.

High-Yield Exam Takeaways & Common Traps

- The cornea contributes 2/3 of total refraction, whereas the lens contributes 1/3.
- Only the lens can change its shape (via accommodation using ciliary muscles) to alter focus distance.
- Images hitting the retina are inverted, meaning the top of the visual field hits the bottom of the retina.

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9. Retinal Architecture, Phototransduction, and the Fovea

The **retina** is a multi-layered neural structure containing five distinct classes of neurons. **Photoreceptors** (rods and cones) embedded within the retina are responsible for **phototransduction**—converting incoming photons of light into electrical signals.

- **Rods**: Highly sensitive to low light levels (scotopic vision) and dominate the peripheral retina; specialized for detecting motion and environmental context.
- **Cones**: Responsible for high-acuity and color vision (photopic vision).
- **Fovea**: A specialized depression in the retina dedicated to fine-detail vision. It contains **exclusively cones** and lacks rods.

To ensure optimal visual acuity, light must reach the photoreceptors with

minimal distortion. The retina lies in front of the **pigment epithelium** (which contains **melanin** to absorb stray light and prevent internal reflections). Furthermore, at the fovea, proximal retinal neurons and their cell bodies are physically **shifted to the side**, allowing light direct, unimpeded access to the cone photoreceptors. Axons of **retinal ganglion cells** converge to form the **optic nerve**, propagating electrical signals to higher visual processing centers.

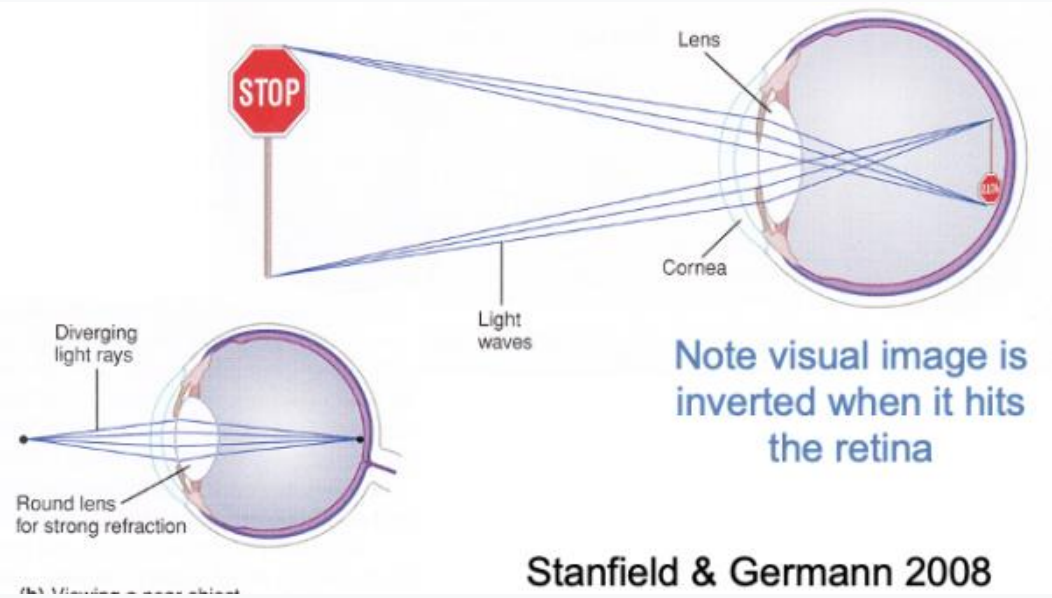


Diagram Walkthrough (slide_06_fig_3.jpeg): Microscopic architecture of the retina showing light refraction onto the fovea and the specialized lateral displacement of proximal retinal circuitry (ganglion and bipolar cells) at the fovea to provide photoreceptors with direct, unobstructed access to light.

High-Yield Exam Takeaways & Common Traps

- The fovea contains exclusively cones and is responsible for high-resolution central vision.
- Melanin within the pigment epithelium absorbs stray light to prevent image degradation from back-reflection.
- Retinal ganglion cell axons bundle together to form the optic nerve exiting the eye at the optic disk.

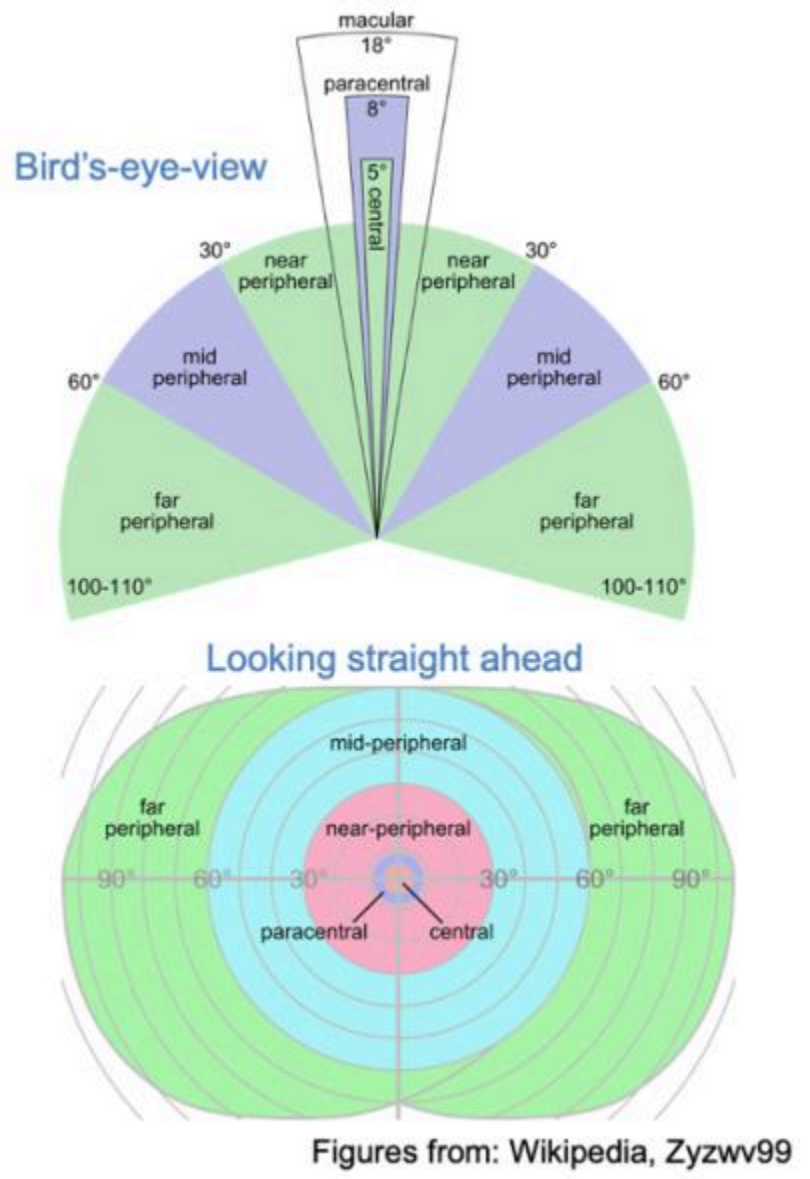


Diagram Walkthrough (slide_07_fig_1.jpeg): Slide 7 figure illustrating the anatomical layout of the retina and the bird's-eye vs. straight-ahead views of the visual field. Note how central vision is concentrated in the paracentral/macular zones (cones), while peripheral vision extends broadly across rod-dominated zones.

High-Yield Exam Takeaways & Common Traps

- Central vision relies almost exclusively on cone photoreceptors and

governs high-acuity object identification.
- Peripheral vision uses predominantly rods to monitor environmental context and moving limbs.
- Retinal ganglion cell bodies in the fovea are displaced laterally to ensure light strikes the foveal photoreceptors with minimal optical distortion.

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11. Visual Fields and Visual Pathways

Light entering the eye projects inverted onto the retina, creating a strict segregation between the nasal hemiretina (closer to the nose) and the temporal hemiretina (closer to the temples).

Step-by-Step Causal Chain of Visual Information Flow:

1. **Retinal Projection**: The visual field is divided into left and right halves. Light from the **left visual field** strikes the **nasal hemiretina of the right eye** and the **temporal hemiretina of the left eye**.

2. **Optic Nerve & Chiasm Decussation**: Retinal ganglion cell axons leave the eye via the optic nerve. At the **optic chiasm**, axons originating from the **nasal hemiretinas cross over** (decussate) to the contralateral side, whereas axons from the **temporal hemiretinas remain uncrossed** (ipsilateral).

3. **Optic Tract Integration**: Because of this partial decussation, the **right optic tract** carries information exclusively from the **left visual field** (from both eyes), and the **left optic tract** carries information exclusively from the **right visual field**.

4. **Thalamic Relays**: Optic tract fibers synapse at the **Lateral Geniculate Nucleus (LGN)** of the thalamus.

5. **Optic Radiations to Cortex**: Neurons from the LGN project via the optic radiations to the **Primary Visual Cortex (V1** / **Brodmann Area 17)** in the

occipital lobe, preserving complete contralateral representation.

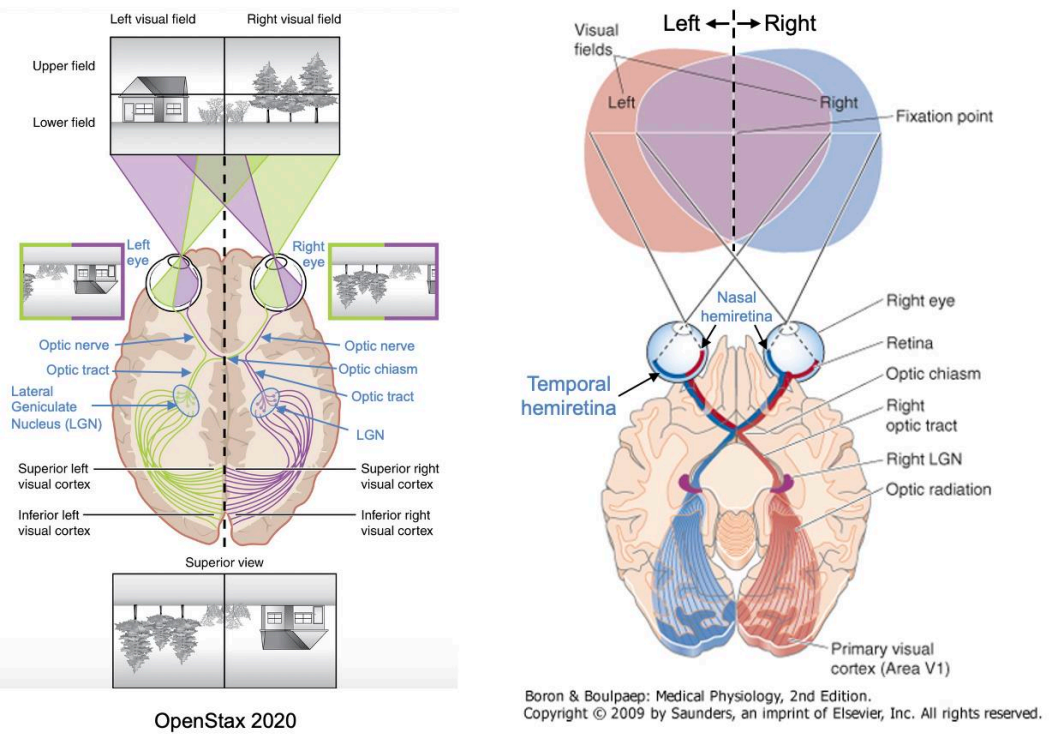


Diagram Walkthrough (slide_08_fig_1.jpeg): Slide 8 top figure illustrating the optics of visual field projection, optic nerve crossover at the optic chiasm, synaptic relay in the LGN, and termination in the superior and inferior banks of the primary visual cortex.

High-Yield Exam Takeaways & Common Traps

- The left visual field is processed entirely by the right cerebral hemisphere, and vice versa.
- Only fibers from the nasal hemiretinas cross at the optic chiasm; temporal hemiretina fibers stay ipsilateral.
- Axons exiting the LGN travel through the optic radiations to terminate in striate cortex (V1).

12. Clinical Lesions of the Visual Pathway

Damage at specific anatomical loci along the visual pathway yields distinct, predictable deficits in visual field perception (scotomas and hemianopsias), serving as a powerful diagnostic tool in neurophysiology:

- **1. Lesion of the Optic Nerve**: Results in a **total loss of vision in one eye** (ipsilateral blindness) because visual input from that eye is completely severed before reaching the optic chiasm.
- **2. Lesion of the Optic Chiasm**: Typically caused by pituitary adenomas pressing on the center of the chiasm. This damages crossing fibers from both nasal hemiretinas, resulting in **bitemporal hemianopsia** (loss of the outer peripheral vision in both eyes, known as 'tunnel vision').
- **3. Lesion of the Optic Tract**: Results in **contralateral homonymous hemianopsia** (loss of the entire opposite half of the visual field in both eyes, e.g., a right optic tract lesion wipes out the entire left visual field).

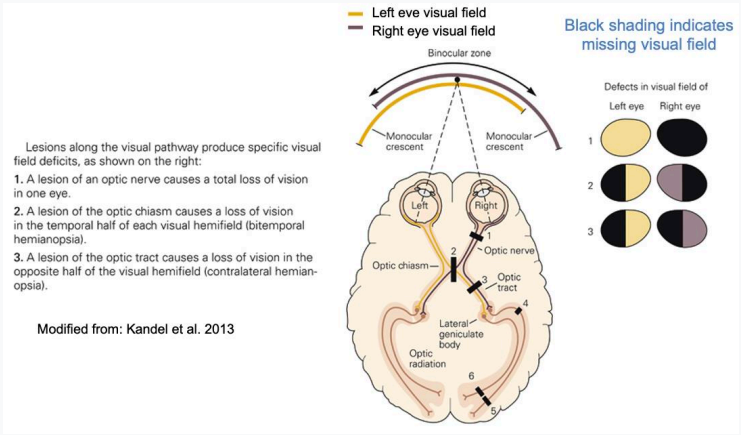


Diagram Walkthrough (slide_08_fig_2.jpeg): Slide 8 bottom figure showing specific visual pathway lesions (numbered 1, 2, and 3) and their corresponding visual field deficits depicted in black shading for the left and right eyes.

High-Yield Exam Takeaways & Common Traps

- Optic nerve damage = ipsilateral monocular blindness (one complete eye lost).
- Optic chiasm damage = bitemporal hemianopsia (loss of outer temporal visual fields).
- Optic tract damage = contralateral homonymous hemianopsia (loss of the entire contralateral visual hemifield).

13. Binocular Retinal Disparity and Depth Perception

Binocular disparity refers to the **difference in image location** of an object seen by the left and right eyes, resulting directly from the eyes' horizontal separation (parallax). When we fixate on a target point, light rays from objects at different depths fall on mismatched (non-corresponding) points on the two retinas. **Integrating this retinal disparity allows the central nervous system to perceive depth and 3D space.** Furthermore, binocular disparity can signal object motion: if an object moves while the eyes remain stable, its image activates the retina progressively more laterally over time. Conversely, when tracking an object, **oculomotor vergence feedback** (derived from muscle spindles in extraocular muscles as the eyes converge or diverge) works alongside disparity to compute distance and time-to-contact.



Diagram Walkthrough (slide_09_fig_1.jpeg): Slide 9, Figure 1: Illustrates how light rays from objects at varying distances (e.g., points A and B relative to a fixation port) strike different retinal locations in the left and right eyes,

generating binocular disparity and depth perception.

High-Yield Exam Takeaways & Common Traps

- Binocular disparity relies on inter-ocular parallax and non-corresponding retinal loci.
- Oculomotor vergence feedback provides proprioceptive signals from extraocular muscle spindles during convergence and divergence.
- Exam trap: Do not confuse monocular cues with binocular disparity, which strictly requires horizontal eye separation.

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14. Anatomy and Retinotopic Organization of the Primary Visual Cortex (V1)

The primary visual cortex (V1), also known as the **striate cortex** or **Brodmann area 17**, is approximately 2 mm thick and contains the standard 6 layers characteristic of neocortex. Afferent inputs carrying visual signals from the lateral geniculate nucleus (LGN) of the thalamus terminate primarily in **layer 4** (specifically layer 4c). The term **'striate'** originates from the prominent white-matter stripe visible in cross-sections, created by dense myelinated axon bundles in layer 4. V1 exhibits strict **retinotopic representation**, meaning that adjacent points in visual space are mapped onto adjacent points across the cortical surface. Furthermore, contralateral visual field representation is fully preserved: light from the left visual field projects to the right cerebral hemisphere, and vice versa.

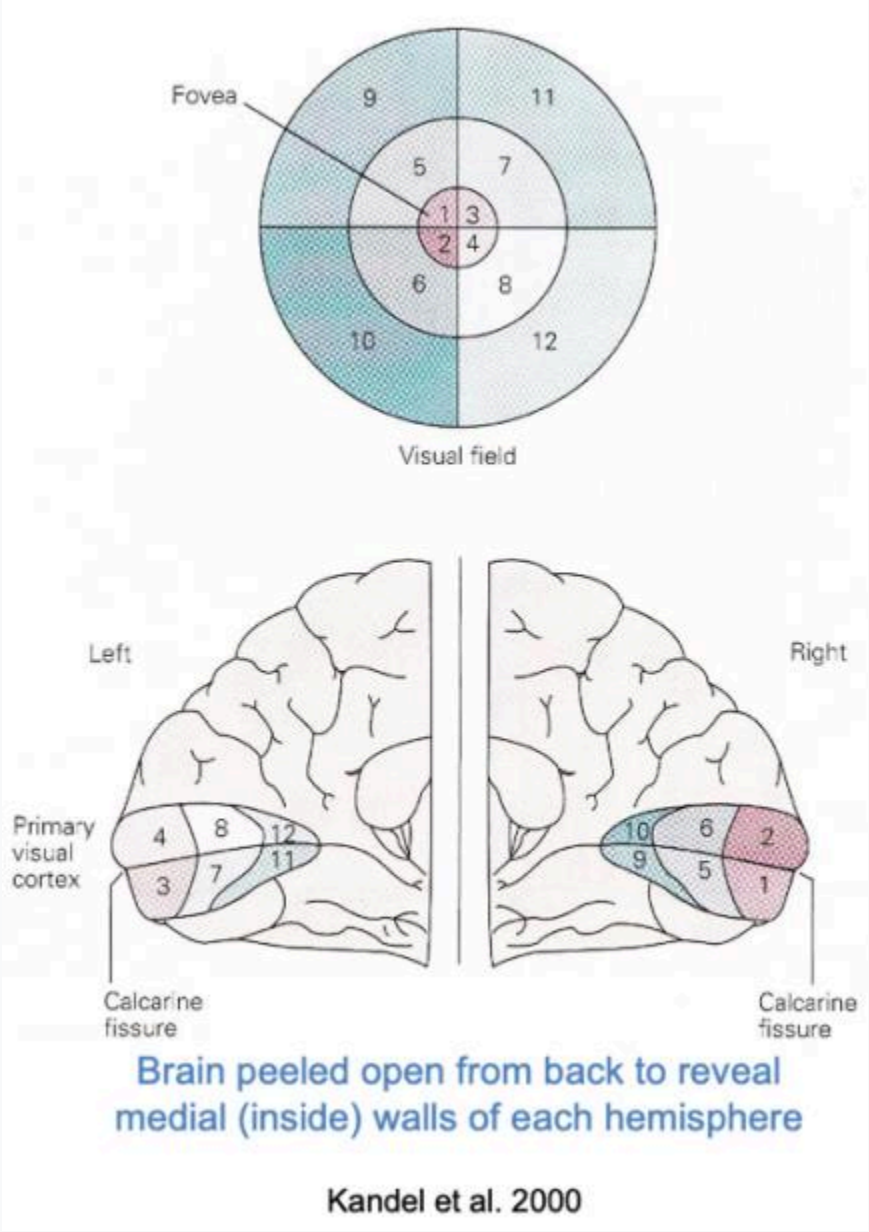


Diagram Walkthrough (slide_09_fig_2.jpeg): Slide 9, Figure 2: Demonstrates retinotopic mapping across the visual field and the medial wall of the occipital lobe surrounding the calcarine fissure (Primary Visual Cortex / V1).

High-Yield Exam Takeaways & Common Traps

- V1 is synonymous with the striate cortex and Brodmann area 17.
- LGN afferents project directly into Layer 4 (Layer 4c).
- Retinotopy ensures spatial adjacency in the visual field is

> preserved topographically in the cortex.

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15. Hierarchical Receptive Field Properties in V1: Layer 4c, Simple, and Complex Cells

Visual processing in V1 follows a strict **hierarchical organization** where receptive fields expand and increase in functional complexity at each synaptic relay:

Cell Type	Anatomical Location	Receptive Field Characteristics	Primary Stimulus Trigger
Retinal Ganglion & LGN Cells	Retina / Thalamus	Tiny, circular spots	Small point of light
Layer 4c Cells	V1 Layer 4c	Small, circular	Tiny spot of light (similar to LGN)

Simple Cells	V1 (Early Layers)	Elongated, rectangular	Bars of light, light/ dark borders, specific orientations
Complex Cells	V1 (Later / Higher Layers)	Larger, expansive	Oriented lines/edges *anywhere* within the field; moving bars

At each hierarchical step, a downstream neuron's receptive field is constructed by summing the convergent inputs of multiple smaller, upstream receptive fields.

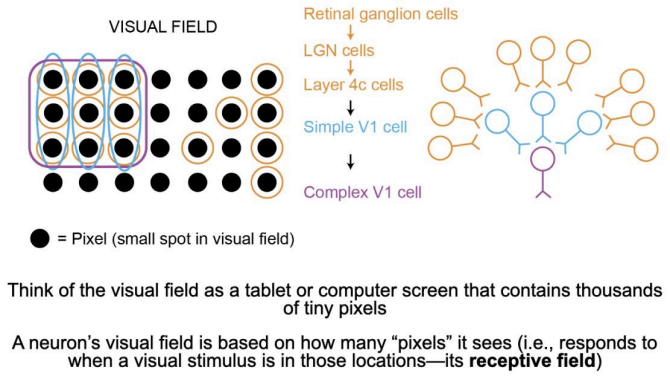


Diagram Walkthrough (slide_10_fig_1.jpeg): Slide 10, Figure 1 & 2: Neural circuitry and hierarchical pooling demonstrating how circular receptive fields of LGN/Layer 4c cells converge to form elongated simple cell receptive fields,

which in turn converge to form complex cell receptive fields.

High-Yield Exam Takeaways & Common Traps

- Simple cells detect static oriented bars and edges; complex cells detect oriented edges moving across a larger receptive field.
- Hierarchical processing is a fundamental rule: downstream receptive fields represent the sum of upstream inputs.
- Exam trap: Complex cells do not require a light bar to be in an exact spatial sub-region; they respond to the correct orientation anywhere within their larger receptive field.

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16. Neuronal Tuning Curves and Visual Feature Selectivity in V1

As visual information ascends from the lateral geniculate nucleus (LGN) into layer 4c of the primary visual cortex (V1, Brodmann area 17), simple and complex cells acquire selectivity for specific stimulus features, most notably the **orientation of lines or edges**. To quantify this selectivity neurophysiologically, researchers record action potential discharge frequencies while presenting bars of light at varying angles across a screen. Plotting firing rate ($ ext{Hz}$) against stimulus orientation yields a **neuronal tuning curve**.

Key features of neuronal tuning curves include:

- **Preferred Orientation**: The specific angle of a stimulus that elicits the maximum discharge frequency (peak of the curve).
- **Bandwidth** / **Selectivity**: The range of orientations that drive the neuron above baseline activity. A **narrow tuning curve** (e.g., the orange neuron in slide figures) indicates high specificity, responding only to a very tight

window of orientations. A **broader tuning curve** indicates lower specificity, responding across a wider range of angles.

- **Population Coding**: Because individual neurons possess broad tuning curves, the precise orientation of an edge is encoded across a **population of neurons** rather than by a single cell alone.

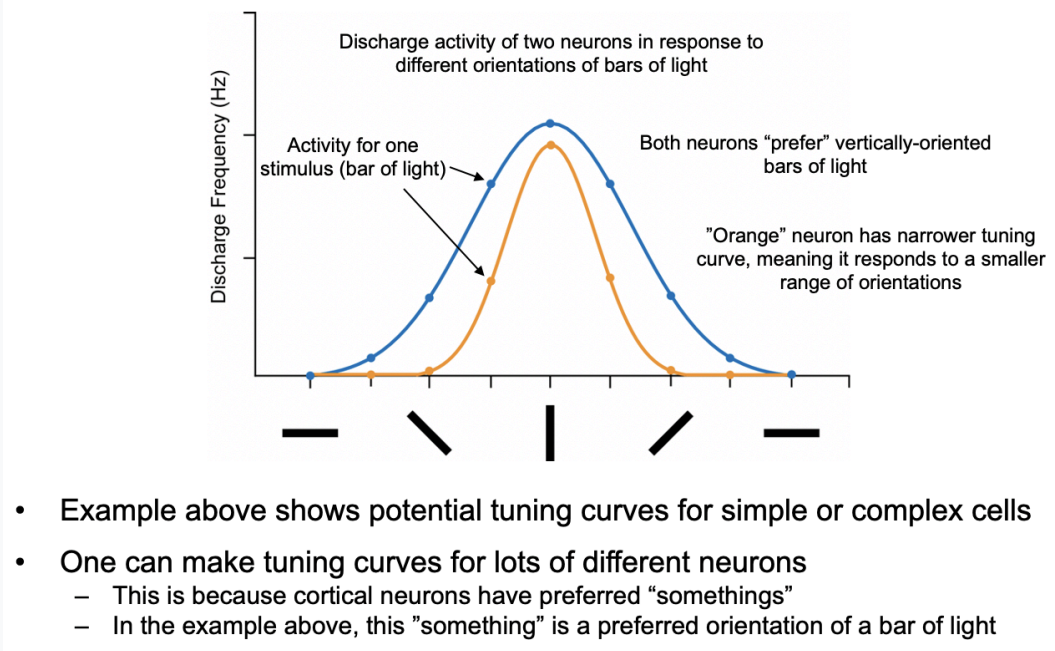


Diagram Walkthrough (slide_11_fig_1.png): Slide 11 Figure 1 illustrates neuronal tuning curves for two cortical neurons responding to oriented bars of light. Observe that both neurons prefer vertical orientations (peak firing rate at 90 degrees), but the orange neuron exhibits a narrower tuning curve, signifying a smaller range of accepted orientations compared to the broader blue curve.

High-Yield Exam Takeaways & Common Traps

- A narrower tuning curve represents higher orientation selectivity (responds to a smaller range of stimuli).
- Single neurons cannot uniquely code exact orientation due to broad firing profiles; the CNS utilizes population coding across

multiple differently-tuned neurons.
- V1 is also known as the striate cortex due to the prominent myelinated layer 4, which gives cross-sections a striped appearance.

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17. Extrastriate Cortical Regions and Visual Motion Processing (MT/V5 and MST)

Beyond the primary visual cortex (V1 / striate cortex), all surrounding visual areas are collectively termed **extrastriate cortex**. Processing visual motion for perception and motor control requires specialized extrastriate regions:

- **Middle Temporal (MT) Region (also designated Area V5)**:
- Contains neurons with **large receptive fields**.
- Exhibits strong **direction preference** (neurons discharge maximally when moving in a specific preferred direction).
- Firing rate is sensitive to the **speed of motion**.
- Projects directly downstream to the MST region.
- **Medial Superior Temporal (MST) Region**:
- Positioned adjacent to MT and characterized by **even larger receptive fields**.
- Divided into distinct sub-regions dedicated to processing **object-motion** (moving objects in the environment) versus **self-motion** (optic flow generated by bodily movement).

Hierarchical integration in the visual pathway operates through convergent feedforward connections: upstream cells with small receptive fields provide inputs that summate to create downstream cells with progressively larger and

more complex receptive fields.

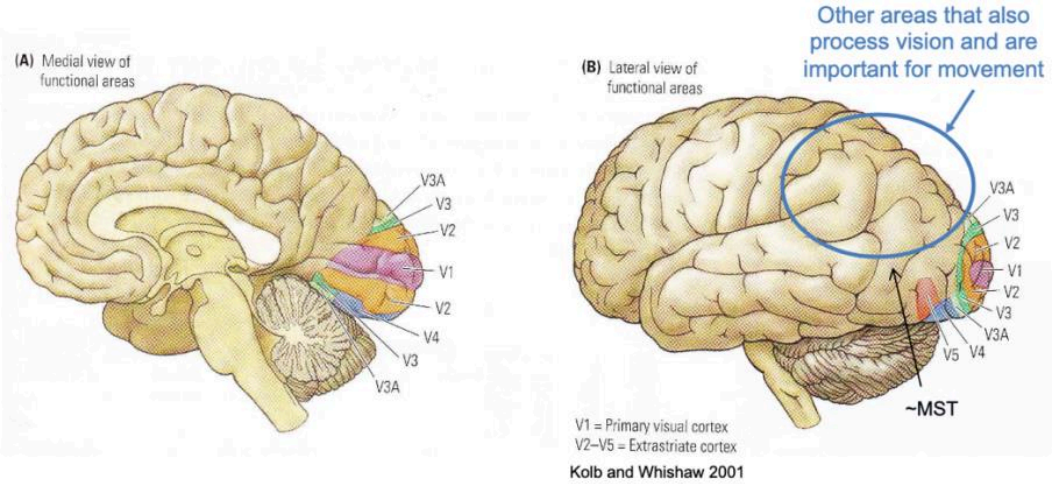


Diagram Walkthrough (slide_12_fig_1.jpeg): Slide 12 displays medial and lateral views of the primate brain highlighting extrastriate areas V2, V3, V4, MT/V5, and MST, alongside a wiring diagram illustrating the staggering complexity of bidirectional hierarchical connections between cortical visual areas.

High-Yield Exam Takeaways & Common Traps

- Area MT (V5) is specialized for visual motion, featuring neurons sensitive to motion direction and speed with large receptive fields.
- Area MST contains even larger receptive fields and is functionally subdivided into processing object-motion and self-motion (optic flow).
- Extrastriate cortex refers to all visual cortical areas outside of V1 (Brodmann area 17).

18. Optic Flow Mechanisms and Spatial Navigation

Optic flow is defined as the continuous change over time of the spatial pattern of light reaching the eye as an individual moves through their surroundings. **Translational flow (radial outflow)** occurs when moving forward with a straight gaze, where visual patterns expand outward from a central focal point. **Rotational flow** is generated when the eyes turn within their sockets. **Key components of optic flow processing include:**

- **Middle Temporal (MT / V5) Region:** Contains neurons with large receptive fields tuned to specific directions and speeds of motion.
- **Medial Superior Temporal (MST) Region:** Integrates MT output into sub-regions responsible for analyzing object motion versus self-motion.
- **Postural Stability & Balance:** Optic flow provides closed-loop feedback regarding postural sway (no sway equals no flow). Disrupting optic flow leads to immediate compensatory sway adjustments.

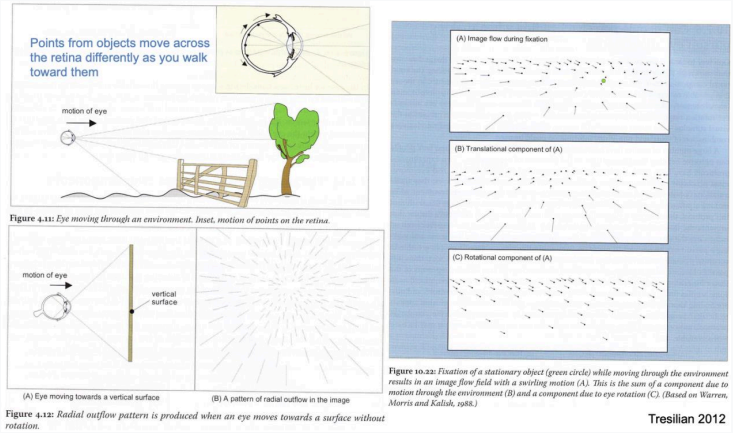


Diagram Walkthrough (slide_13_fig_1.jpeg): Slide 13 illustrates optic flow components. Panel A shows global image motion during fixation, Panel B depicts pure translational radial outflow during forward movement, and

Panel C demonstrates rotational flow components due to eye rotation.

High-Yield Exam Takeaways & Common Traps

- Remember that closer objects sweep across the retina faster than distant objects during translational flow.
- Optic flow is essential for determining heading direction, velocity, and stability during locomotion.
- Area MT (V5) and MST are critical extrastriate hubs for processing visual motion.

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19. Time-to-Contact (TTC) Estimation

Time-to-contact (TTC) serves a critical predictive function for movement initiation, obstacle avoidance, and interceptive behaviors. **The three primary sensory mechanisms for calculating TTC are:**

Mechanism	Physiological / Mathematical Basis	Functional Role
Optic Flow (Tau Variable)	Ratio of retinal image size to rate of radial expansion	Rapid predictive calculation of impending collision

Binocular Retinal Disparity	Horizontal separation parallax between left and right retinas	Depth and distance perception
Oculomotor Vergence Feedback	Muscle spindle feedback from extraocular muscles during convergence/ divergence	Tracking objects moving toward or away from the face

High-Yield Exam Takeaways & Common Traps

- Tau is the specific optic flow variable defined as the retinal image size divided by its rate of expansion.
- Oculomotor vergence relies directly on proprioceptive feedback from extraocular muscle spindles.

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20. Dorsal and Ventral Visual Streams (The Two Visual Systems)

The cerebral cortex processes visual information along two major divergent pathways following primary visual cortex (V1) processing:

- **Ventral Visual Stream ('What' Pathway):** Extends from V1 to the temporal lobe. It is responsible for fine visual analysis (form, color, object recognition) and utilizes an object-centered frame of reference.
- **Dorsal Visual Stream ('How' / 'Where' Pathway):** Extends from V1 to the posterior parietal cortex and beyond. It is dedicated to guiding goal-directed actions and spatial interactions, utilizing egocentric (body-based)

frames of reference.

Double Dissociation Evidence: Experiments involving brain lesion patients demonstrate independence between streams. Patient **D.F.** (ventral damage) cannot visually discriminate shapes but can successfully grasp objects. Patient **R.V.** (dorsal damage) can discriminate shapes but exhibits improper finger grips during object manipulation.

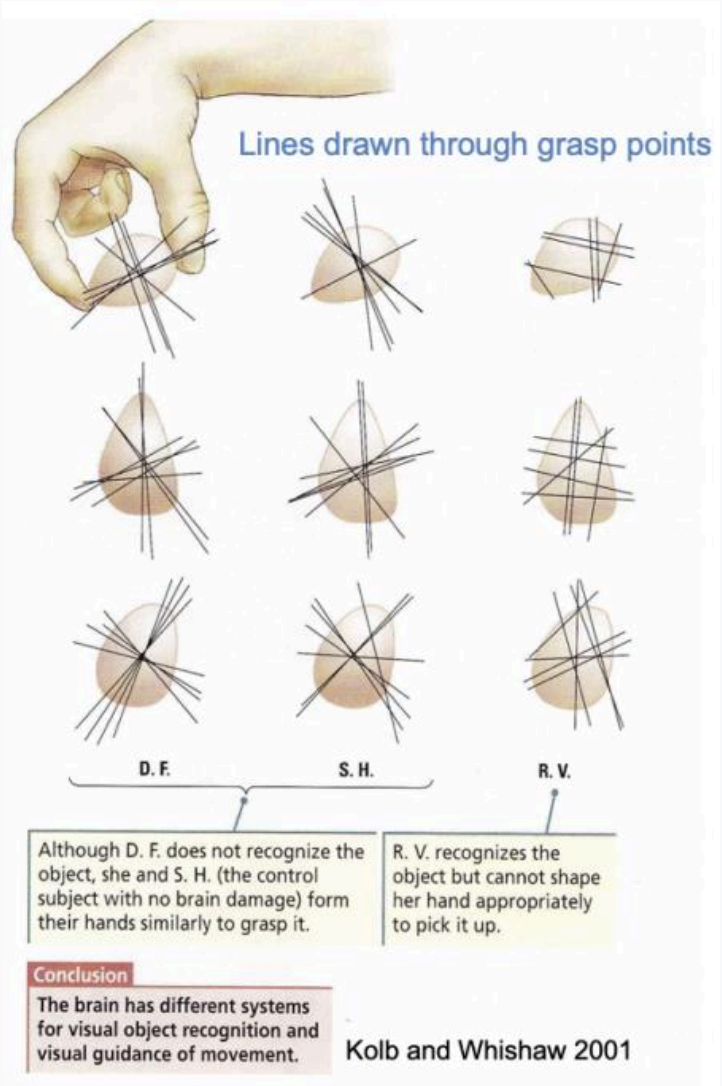


Diagram Walkthrough (slide_15_fig_1.jpeg): Slide 15 illustrates the anatomical segregation of the ventral ('what') and dorsal ('how/where') visual streams in the human cortex.

> **High-Yield Exam Takeaways & Common Traps**
>
> - Ventral stream = temporal lobe = 'what' = perception & object recognition.
> - Dorsal stream = parietal lobe = 'how/where' = action, reaching, & grasping.
> - A double dissociation provides definitive scientific proof that two cognitive functions operate via separate neural pathways.

21. Vision and Standing Balance: The Moving Room Paradigm

Vision plays an indispensable role in the maintenance of upright posture and balance. When somatosensory and vestibular inputs provide ambiguous or conflicting signals, the central nervous system (CNS) frequently assigns **visual primacy**—giving overriding weight to visual cues.

The Moving Room Experiment (Lee & Aronson 1974):

1. **Experimental Setup:** Adult participants or toddlers stood inside a specially constructed room where the walls and ceiling could be translated forward or backward, while the floor remained strictly stationary.

2. **Sensory Conflict:** Movement of the surrounding walls generated a massive visual perception of self-motion (vection), even though the somatosensory system registered no ankle angle changes and the vestibular system detected no true acceleration.

3. **Behavioral Output:** Participants initiated immediate, unnecessary postural sway and corrective adjustments in the direction opposite to the wall movement. Infants unaccustomed to independent standing invariably lost their balance and fell over, demonstrating the immature reliance on visual feedback for postural

equilibrium.

Feature	Moving Room Experiment	Sensory Organization Test (SOT)
Primary Manipulation	Moving walls/ ceiling relative to stationary floor	Sway-referenced visual surround and/or platform
Sensory Conflict Type	Visual vs. Somatosensory / Vestibular	Visual vs. Somatosensory conflict via altered reference frames
Behavioral Response	Unnecessary postural sway or loss of balance (falls in infants)	Increased body sway and quantification of sensory weighting

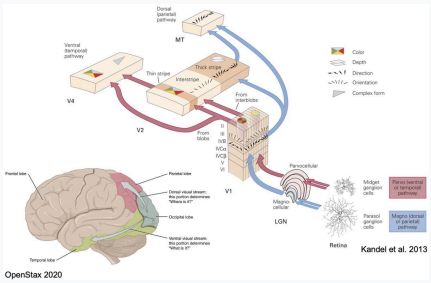


Diagram Walkthrough (slide_16_fig_1.jpeg): The moving room apparatus (Lee & Aronson 1974). Translating the walls and ceiling forward or backward creates conflicting visual optic flow that overpowers somatosensory and vestibular

feedback, inducing robust postural sway.

High-Yield Exam Takeaways & Common Traps

- Adults and infants give primacy to visual information when visual and somatosensory cues conflict, leading to compensatory sway during the moving room experiment.
- The moving room specifically translates walls and ceiling while keeping the floor stationary.

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22. Visual Guidance of Locomotion and Foot Placement

Vision is essential for planning, navigating, and executing safe locomotor trajectories across complex terrains. Rather than operating purely as a feedforward map, visual feedback provides continuous online guidance.

Key Contributions of Vision to Walking:

- **Obstacle Avoidance & Terrain Accommodation:** Visual cues allow pedestrians to identify curbs, uneven surfaces, and obstacles well in advance, implementing targeted avoidance strategies.
- **Heading Direction & Navigation:** Peripheral and central optic flow fields inform the CNS of the current walking trajectory and speed.
- **Gaze Fixation & Look-Ahead Distance:** Individuals naturally fixate on exact spots where they intend to step. The **look-ahead distance** dynamically decreases as terrain complexity increases (requiring tighter spatial precision).
- **Online Trajectory Monitoring:** Vision tracks lower limb position relative to obstacles (e.g., when stepping over a barrier) to update motor commands

on-the-fly.

- **Dual Role:** Vision is utilized both *proactively* to plan upcoming movement phases and *reactively* as sensory feedback to correct unfolding motor patterns.

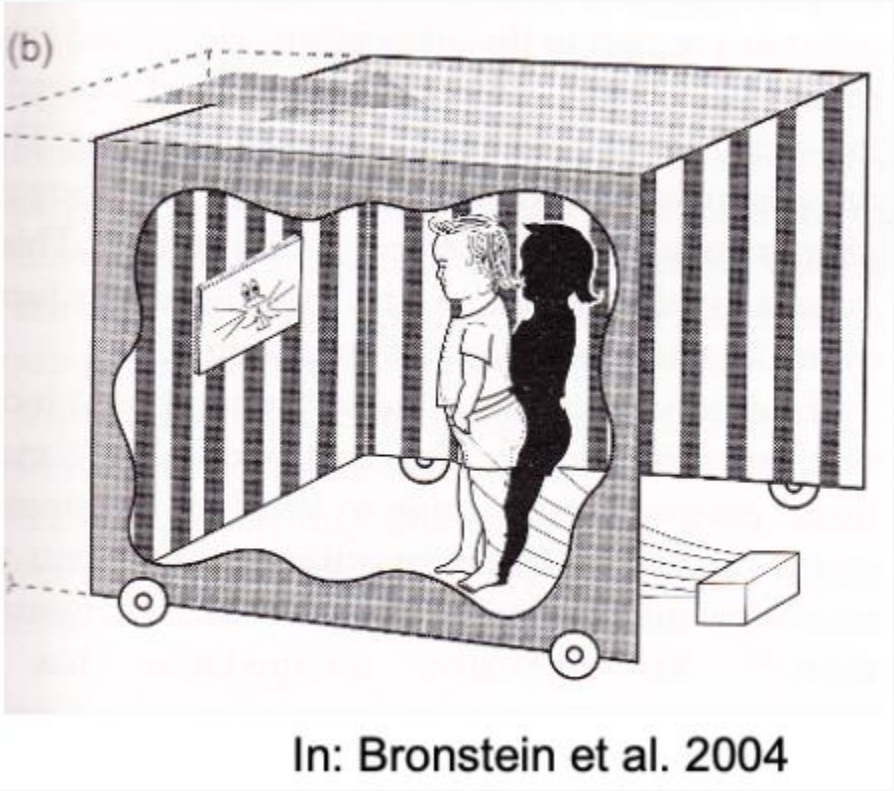


Diagram Walkthrough (slide_16_fig_2.jpeg): Summary of visual contributions to locomotion, highlighting avoidance strategies, look-ahead fixation on complex terrain, and online limb trajectory monitoring.

High-Yield Exam Takeaways & Common Traps

- Look-ahead distance during locomotion decreases as the complexity of the terrain increases.
- Vision serves both a planning (feedforward) role and an online correction (feedback) role during human locomotion.

23. Visual Contributions to Reaching and Movement Kinematics

Reaching out to grasp or interact with objects relies heavily on vision to establish spatial coordinates and monitor limb kinematics.

Mechanisms of Visual Control in Reaching:

- **Extrinsic World-Based Coordinates:** Vision provides an external reference frame used by the dorsal stream ('how'/'where' pathway) to plan spatial features of movements toward visual targets.
- **Initial Limb Configuration:** Vision provides immediate feedback regarding the starting position of the arm and hand relative to the target.
- **Consequences of Vision Deprivation:** Complete absence of vision of the hand and arm during reaching results in distinct kinematic alterations, including **longer movement times** and **decreased endpoint accuracy**.
- **Continuous vs. Intermittent Feedback:** Experimental evidence confirms that visual feedback is integrated continuously throughout the movement, particularly after the initial feedforward ballistic phase.

> **High-Yield Exam Takeaways & Common Traps**
>
> - Removing vision of the arm during reaching leads to longer movement durations and decreased spatial accuracy.
> - Visual feedback during reaching is utilized continuously throughout the unfolding movement, not just at movement onset.